

CreditWise Loan Approval Prediction Project

Instruction Manual (Competition-Based Evaluation)

Faculty: Dr. Jilan Samiuddin
Teaching Assistant: Bijoy Deb Babu
Course: CSE 445 (Machine Learning), Section 8

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Abstract

This document describes the CreditWise Loan Approval Prediction Project. Students will build a complete machine learning pipeline on a tabular dataset to predict whether a loan should be approved (`Loan_Approved=1`) or rejected (`Loan_Approved=0`). The evaluation is competition-based. **Model ranking will be based on the F1-score** (higher is better). Final grading is based on the notebook (60 marks) and the written report (40 marks).

1 Project Overview

You are provided with a loan application dataset containing applicant demographic, financial, and credit-related features. Your task is to train **a minimum of 3** machine learning models to predict the target label:

- **Target:** `Loan_Approved` (1 = Approved, 0 = Rejected)

2 Project Objectives

By completing this project, you will be able to:

- Identify and handle missing values in real-world tabular data with proper justification.
- Perform exploratory data analysis (EDA) to understand class balance, distributions, and potential outliers.
- Apply preprocessing steps including categorical encoding, multicollinearity analysis, and scaling where appropriate.
- Train and evaluate multiple classification models using consistent metrics and visual diagnostics.
- Compare models and justify which model performs best using both quantitative and qualitative reasoning.
- Apply feature engineering and evaluate whether performance improves, supported by comparisons.
- Write a clear and reproducible technical report.

3 Expected Deliverables

- **Source Python Notebook** (`.ipynb`) containing all required outputs and diagrams (plots, tables, metrics, confusion matrices).
- **Report** in PDF format (`.pdf`), 5–12 pages (including references/appendix).
- **Submission format:** Upload both files together in a single `.zip` file (**`.rar` is not allowed**).

4 Notebook Marking Rubric (Total: 60 Marks)

The notebook will be graded using the following criteria:

1. Data Cleaning (10 marks)

- Show missing value counts (per column + total).
- Handle missing values using a justified method (drop / impute).
- Show evidence of the final cleaned dataset (e.g., summary after cleaning).

2. Exploratory Data Analysis (EDA) (10 marks)

- Show class distribution of `Loan_Approved` and comment on balance.
- Check numerical features for outliers (only report meaningful findings).
- Plot and explain a histogram showing `Credit_Score` vs `Loan_Approved`.

3. Preprocessing Pipeline (14 marks)

- Apply correct encoding for categorical features (e.g., One-Hot / Ordinal) and justify the choice.
- Apply feature scaling where appropriate; justify why scaling is needed or not needed for your selected models.
- Ensure no data leakage: fit encoders/scalers on training data only.
- Clean implementation is expected.

4. Correlation & Multicollinearity (8 marks)

- Plot a correlation heatmap for numerical features.
- Identify highly correlated feature pairs (multicollinearity).
- If multicollinearity exists, apply a corrective step (drop/merge) and justify briefly.

5. Model Training, Evaluation & Best-Model Justification (18 marks)

- Perform train/test split (recommended 80/20).
- Train and evaluate at least **three** different classification models.
- For each model, report: Accuracy, Precision, Recall, **F1-score**.
- For each model, show a confusion matrix heatmap using seaborn.
- Select the best model based primarily on **F1-score** and justify the selection using results.

5 Report Marking Rubric (Total: 40 Marks)

The written report will be graded using the following criteria:

1. Problem Statement & Goal (8 marks)

Clear description of the task, the target variable, and the objective of the project.

2. Dataset Overview & Key Observations (8 marks)

Description of features/target and key insights from the dataset (e.g., class balance, important trends).

3. Methodology (8 marks)

Explanation of decisions and workflow: missing value strategy, preprocessing choices (encoding/scaling), and how the models were trained. Justification is required (brief but clear).

4. Results Presentation (8 marks)

Clear presentation of results for all three models including a comparison of metrics and confusion matrices (tables/figures are expected).

5. Discussion & Best Model (F1-score) (8 marks)

Reasoned conclusion on which model performed best based on F1-score, supported by evidence and explanation of errors/behavior.

6 Grading Scheme (Total: 100 Marks)

Your final grade for this project will be calculated as follows:

Component	Marks
Notebook submission (.ipynb)	60
Report submission (.pdf)	40
Total	100

7 General Rules

- **Reproducibility:** Keep your notebook runnable end-to-end. The notebook will be executed in colab to check if the notebook can produce outputs reported.
- **No Data Leakage:** Perform train/test split before fitting encoders/scalers.
- **Clarity:** Use clear headings and write short, precise explanations for each decision in the Mark-down of the notebook.
- **Academic Integrity:** Any external code or resources used must be cited in the report.